UVA | SCHOOL of DATA SCIENCE

DS 2003: Communicating with Data
Fall 2026

Course Information

Course Start Date–End Date: Aug 25th – Dec 8, 2026
Credit Hours: 3
Meeting Days & Times: Mo|We|Fr 12:00 PM – 12:50 PM
Location: Data Science Building Room 305
Catalog Description: The course is designed to not only teach students the tools necessary to visualize data but also effective techniques for explaining data-driven results with an emphasis on communicating statistical output in a manner that best represents the findings. Examples might include tailoring messages based on the audience or shaping visualizations to follow a storyline. Content on the development of interactive plots and dashboards will also be included.
Requisites: Students must have completed CS 1110 or CS 1111 or CS 1112 or CS 2110 or DS 1002 or PHYS 1655 or CS 1113

Instructor Information

Sarah Groves, Assistant Professor SDS
Email: smgroves@virginia.edu
Office Hours: Fridays 1:00 pm – 2:00 pm
Office Location: Rm 437

Instructional Support: Isabelle Lee
Email: vjm3mp@virginia.edu
Office Hours: Wednesdays 1:00-2:00pm
Office Location: Capitol One Hub

Learning Outcomes

By the end of the semester, you will be able to:

- Understand the importance of **data communication** in various domains.
- Gain knowledge about various tools and techniques, including proficiency in using **industry-standard software for data visualization and reporting**.
- Apply basic **design principles** to create visually effective and informative data visualizations.
- Develop **storytelling** arches to richly describe data alongside visualizations.
- Analyze how charts can **misrepresent data**.
- Critique the **effectiveness** of various data communication strategies and make informed decisions about which techniques are best suited for specific communication objectives and target audiences.

Semester Schedule

Refer to the Canvas calendar tool for specific dates and deadlines. All changes to due dates will be reflected in Canvas.

Wk	Dates	Topic	Monday	Wednesday	Friday	Deadlines
1	*Aug 24 -28*	**Introduction**	—	Introduction	Lab 0: Environment setup	
2	*Aug 31 – Sep 4*	**Know your material**	Why we visualize	Data Types, Tables, and Models	Lab 1: Intro to pandas and data frames	P1 Assigned
3	*Sep 9 & 11*	**Visualization encodings**	Labor Day (no class)	Encodings	Lab 2: Grouping and aggregating	P1 topic and dataset picked
4	*Sep 14 – 18*	**Effective visualization**	Expressiveness and effectiveness	Scales and colors	Lab 3: Anatomy of a figure	
5	*Sep 21 – 25*	**Perception**	How perception works	Gestalt principles	Lab 4: More EDA - Encodings, Color, and Chart Types	
6	*Sep 28 – Oct 2*	**Visualization Tasks**	Task Abstractions	Applications	Lab 5: Bad Chart Redesign	
7	*Oct 5 – 9*	**Statistical plots**	Fall break, no class	Statistical plots and uncertainty	Lab 6: Statistical plots	
8	*Oct 12 – 16*	**EDA**	P1 Presentations	P1 Presentations	P1 Presentations	P1 Due
9	*Oct 19 – 23*	**Storytelling & interaction**	Storytelling and narrative structures	Interaction: why and when (taxonomies)	No class (Family weekend) Async Lab 7: Interactive EDA with Plotly	P2 introduced Groups (Wed) Proposal (Sun)
10	*Oct 26 – 30*	**Interactivity & beyond tables**	Interactive notebooks with ipywidgets	Graphs, networks, and trees	Lab 8: ipywidgets	
11	*Nov 2 – 6*	**Infographics**	Infographics	Infographics II: principles of design	Lab 9: Interactive treemaps and sunbursts	
12	*Nov 9 – 13*	**Applications of storytelling**	Science visualizations	Data journalism	Lab 10 (light): Planning your final piece	P2 narrative (Sun)
13	*Nov 16 – 20*	**Lies, rhetoric & critique**	Data quality and deceptive plots	Ethics and rhetoric	P2 design critique	P2 sketches Reflection (Sun)
14	*Nov 23*	**Project work**	P2 work session	No Class (Thanksgiving)	No Class (Thanksgiving)	
15	*Nov 30 – Dec 4*	**Final Presentations**	TBD	Project Presentations (Day 1)	Project Presentations (Day 2)	P2 slides (Tue)
16	*Dec 7*	**Wrap-up**	Last class: Worst Chart Contest	___	____	P2 final due

Course Material

All course materials will be made available on Canvas. Announcements will also be posted on Canvas, and you will receive an email for urgent announcements. For discussions and questions, you are welcome to use Piazza through the Canvas course site.

Textbook/Resources

There is no required textbook for this course. However, I recommend getting a copy of Visualization Analysis and Design, Tamara Munzner, CRC Press, 2014. (http://www.cs.ubc.ca/~tmm/vadbook/). I have a copy in my office if you'd like to reference anything, or you can find more materials online (Talk to me before spending money on a textbook, please ☺).

Course Deliverables

Quizzes: Weekly quizzes on Wednesday on material from the week before. You must be in class to take the quiz; if you have an excused absence, it is your responsibility to email me to set up a time to take it. The lowest two quiz grades will be dropped.

Labs: In-class lab exercises will be turned in for points. Labs give students an opportunity to iterate on the class content with peers.

Projects: The course is project-oriented. It includes two projects designed to provide the stepping stones needed to complete the final project. Tentative due dates for these projects can be found above in this syllabus under the 'Schedule' heading.

- **Project 1 – Exploratory Data Analysis (Individual):** In this project, you will identify a dataset of interest and perform an exploratory analysis to investigate research questions and develop preliminary insights. Your final submission will take the form of a report consisting of captioned visualizations that convey key insights from your analysis, and a lightning talk on your results.
- **Project 2 – Interactive Notebook + Infographic/Science Visualization (Group):** For this project, your group will choose a dataset, explore it in an interactive Jupyter notebook built with Plotly and ipywidgets, and then distill your main finding into a standalone infographic (for a general audience) or science visualization (for a specialist audience). Each team will present their notebook and a draft of their final piece in class at the end of the semester, then revise the final piece based on feedback.

Grade Breakdown

Assignment	Percentage of Overall Grade
Labs	30%
Quizzes (drop lowest 2)	10%
Attendance/Participation	20%
P1: Exploratory Data Analysis	20%
P2: Interactive Notebook + Infographic/Science Visualization	20%

Grading Policies & Expectations

Expectations

- **Show up.** It is really true that half of the work is just showing up! If you show up to class, you are setting aside time to engage with the new material that you otherwise will need to cram, which has been proven again and again ineffective for learning. Also, I want you here!
- ***Engage.*** As you have probably heard from every teacher before, you get out of class as much as you put into it.
- **Trust** that the course structure and assessments are **intentional for your learning** but **ask me for my reasoning if it isn't clear!** My goal is for you to learn new concepts and encode them in your long-term memory, so that in any career (data scientist or otherwise!), you can both critically interpret visualizations made by others *and* make effective and expressive visualizations of data. If you are not being challenged, you can't learn effectively—let me know if this is the case! On the flip side, if the material is feeling exceedingly difficult, I want to know that too. Everyone feels like an imposter sometimes, but feeling like one doesn't mean you are one.
- **Ask questions** and reach out for help when you need it. Come see me in office hours to ask questions, get help on assignments, or just to chat about anything!
- Give me **feedback** about how your learning is going and how we can help
- Be **patient** with me (and yourself!) as I teach this course for the first time and you learn this material for the first time!

Late Work Policy

You can turn in your assignments **up to two days late**. However, for each day that a project is turned in late, we will deduct 10% off the total possible points. That is, one day late is 10% off (i.e. submitting on a Monday something due Sunday at midnight); two days is 20% off. If your assignment is two days late, the maximum number of points (out of 100) that you can receive is 80.

If you need an extension on an assignment or have any extenuating circumstances that prevent you from completing the assignment, please **email both me and the IA** *as soon as possible,* preferably before the assignment deadline if possible. Extensions will be granted on a case-by-case basis.

Class Attendance & participation

Your **participation** will be measured by your **attendance** and **engagement** in lecture and studio sessions. You are expected to attend each face-to-face class meeting, and 20% of your grade is based on participating in in-class activities. You also will not be able to take the quiz if you are not in class, but this can be one of two dropped quiz grades, or if you know ahead of time that you will need to miss a quiz grade, we can coordinate a time for you to take it early.

Absence Policy

If you are sick, have a religious/sports exemption, or have an extenuating circumstance, please *do not force yourself to come to class;* email me *as soon as possible* for an **excused absence**. Any absence without such an email will be considered unexcused; **excused absences *will not* count against your participation grade.** If you need to be absent for a long time, please email me as soon as is reasonable and we can discuss on a case-by-case basis. For example, a planned trip should be discussed well in

advance of the travel. Please note that I will *not* require you to provide proof of sickness or any details that you do not wish to share with me – I am trusting that, on honor code, your excuse for missing class is legitimate. If serial absences become an issue, we will meet to resolve any problems.

Extenuating Circumstances

If you need help, you are falling behind and struggling to catch up, or if extenuating circumstances arise, the sooner you reach out for help, the more options there are to help and support you. Please contact Heather Corley, Student Success Advisor, at hcorley@virginia.eduif you are experiencing extenuating circumstances, or Beth Wright, a licensed psychotherapist within SDS, at mdj7wf@virginia.edu. See more information at the end of this syllabus.

Grade Scale

The academic performance of a student in each course taken for a grade is recorded as one of the following grades: A+, A, A-; B+, B, B-; C+, C, C-; D+, D, D-; F.

You will be graded according to the following markings:
Grading Scale:

- 97-100 A+
- 93-96.9 A
- 90-92.9 A-
- 87-89.9 B+
- 83-86.9 B
- 80-82.9 B-
- 77-79.9 C+
- 73-76.9 C
- 70-72.9 C-
- 67-69.9 D+
- 63-66.9 D
- 60-62.9 D-
- <60 F

Communication and Student Response Time

Questions containing personal information should be emailed to me. Throughout our time together, the sooner you inform me of any problem that may affect your attendance or performance, the better the chance we have of solving it together.

Academic Integrity: Collaboration and Cheating

Cheating tends to happen at higher rates in introductory programming-based courses because students get frustrated when their code won't run, because of the feeling that there is only one correct way to write the code, and because of how easy it is to copy and paste a few lines of someone else's code.

Although every student is responsible for their own lab reports, you may chat and Zoom with one another to work together on labs. In that context, the line between collaboration and cheating can get a bit fuzzy. In general, the difference between collaboration and cheating comes down to intent. Cheating is trying to circumvent the learning process. Collaboration is trying to help yourself and your classmates learn the material more deeply. Use your own sense of right and wrong, but to help clarify the difference, here are **examples of cheating**:

1. Directly copying someone else's text or code word-for-word, or copying text from ChatGPT
2. Sharing/showing code for the purpose of circumventing the learning process (for example, letting someone copy code because they are running up against the deadline)
3. Asking for help without doing anything to try to solve the problem first; asking someone to do the work for you
4. Sharing answers to reading quizzes

Here are things that are okay:

1. Two people with the same code is okay as long as one person didn't copy-and-paste it from another person. If this is the case, it is preferred that you write the names of anyone you collaborated with at the top of your assignment as a citation of effort.
2. Sharing/showing individual lines of code for the purpose of teaching/explaining or helping someone understand the material
3. Debugging together (this is only possible if both people have already written their own code, otherwise there's nothing to debug)
4. Sharing strategies, external texts, blogs, and other resources for completing problems on the lab assignments

SDS Guidelines on AI Tools and Assistance

The use of generative AI tools and foundation models, (i.e. ChatGPT GPT, DALL-E, Stable Diffusion, Midjourney, GitHub Copilot, and similar tools) is permitted with the following activities in accordance with the stated guidelines at no penalty:

- Brainstorming and refining your ideas
- Fine-tuning your research questions
- Finding information on your topic
- Drafting an outline to organize your thoughts
- Checking grammar and style.

Students are responsible for

- Acknowledging that large language models tend to produce incorrect facts and fake citations.
- Acknowledging that code generation models may produce inaccurate outputs.
- Acknowledging that image generation models can occasionally come up with highly offensive products.
- Taking responsibility for any inaccurate, biased, offensive, or otherwise unethical content submitted, regardless of the origin (i.e., student-generated or from a foundation model).

- Properly citing the contribution of the foundation model or other AI tools in submitted material.
- The entirety of any information they submit, based on an AI query or AI assistance.

The use of generative AI tools is NOT permitted for the following activities:

- Impersonating students in classroom contexts, such as by using the tool to compose discussion board prompts.
- Completing group work assigned to a student, unless it is mutually agreed upon that they may utilize the tool.
- Writing a draft of a writing assignment.
- Writing entire sentences, paragraphs, or papers to complete class assignments.

Students may be penalized for

- Using a foundation model without including an acknowledgement.
- Improperly citing the use of work by other human beings or the submission of work by other human beings as that of the student.
- Violating intellectual property laws.
- Submitting materials containing misinformation or unethical content.

The usage of AI tools must be properly cited to stay within university policies on academic honesty. Failure to adhere to these guidelines will result in a failing grade on the assignment or exam (a zero) and may be an honor code violation depending on the context (to be determined at the instructor's discretion). Having said all these disclaimers, the use of foundation models is encouraged, as it may make it possible for you to submit assignments with higher quality, in less time.

Technical Support Contact

- Login/Password: https://in.virginia.edu/helpdesk
- Canvas: Check the Help tab for resources

School of Data Science Support and Policies

Office of Student Support
The Office of Student Support is here to support your academic journey toward success. Their office provides resources to help you be successful and engaged. If you need support resources related to student success or personal well-being, please reach out to the Office of Student Support directly at: SDS_OSS@virginia.edu.

Student Success Advisor Heather Corley serves as your academic advisor for the duration of your matriculation. Heather provides orientation and clarity on degree requirements, degree planning, course enrollment, transfer credit, institutional policies, and the Student Information System (SIS) and Stellic. Heather is able to help you through personal needs and can connect you with resources across Grounds. Email hcorley@virginia.edu with questions or log into Stellic to make an appointment.

The Data Science Student Portal has resources and information about career, academic support, engagement opportunities, funding, and more.

Career Services
The School of Data Science Career Services Team provides a wealth of opportunities for you to learn, connect, and grow. These offerings are adapted to the needs of the cohort or the year and may take different forms as needs change.

Complete your profile and career interests, explore the Data Analytics Resource Card or make an appointment with Career Services at the School of Data Science by using your UVA Email to Login to Handshake.

Graduate Record
Visit the School of Data Science Graduate Record for policies and information about academic regulations, academic standing, financial assistance, and grades.

University Policies

Academic Integrity and University of Virginia Honor System
The School of Data Science relies upon and cherishes its community of trust. We firmly endorse, uphold, and embrace the University's Honor principle that students will not lie, cheat, or steal, nor shall they tolerate those who do. We recognize that even one honor infraction can destroy an exemplary reputation that has taken years to build. Acting in a manner consistent with the principles of honor will benefit every member of the community both while enrolled in the School of Data Science and in the future. Students are expected to be familiar with the university honor code, including the section on academic fraud.

All work should be pledged in the spirit of the Honor System of the University of Virginia. The instructor will indicate which assignments and activities are to be done individually and which permit collaboration. Students who submit exams electronically acknowledge the Honor Pledge by agreeing to the following statement:

“On my honor, I have neither given nor received aid on this examination, nor did I have prior knowledge of its contents.”

For more information, please visit UVA Honor Committee.

Course Evaluations: Student feedback is critical to the school, the instructor, and future students. Students are expected to complete anonymous and confidential course evaluations in a timely manner for each course at the end of each term.

Discrimination/Harassment/Retaliation
UVA prohibits discrimination and harassment based on age, color, disability, family medical or genetic information, gender identity or expression, marital status, military status (which includes active duty service members, reserve service members, and dependents), national or ethnic origin, political affiliation, pregnancy (including childbirth and related conditions), race, religion, sex, sexual orientation, veteran status. UVA policy also prohibits retaliation. All faculty and TAs are also responsible employees for disclosures or reports of potential discrimination, harassment, and retaliation.

Disability and Pregnancy Accommodations
If you anticipate or experience any barriers to learning in this course, please discuss your concerns with me. If you have a disability, or think you may have a disability, contact the Student Disability Access Center ("SDAC") to request reasonable accommodation(s) for this course. If you have accommodations through SDAC, send me your Faculty Notification Letter as soon as possible and meet with me so we can develop an implementation plan together.

Students may be entitled to reasonable accommodations for pregnancy, childbirth, or related medical issues. Please contact SDAC for additional information. Pregnant and parenting students are encouraged to contact SDAC or EOCR to discuss plans and ensure ongoing access to their academic courses and program. Information for pregnant and parenting students is also available on EOCR's Pregnancy and Parenting Resources webpage.

Religious Academic Accommodations
UVA also provides reasonable accommodations when a student's sincerely held religious beliefs or observances conflict with academic requirements. Students who wish to request an academic accommodation for a religious observance should submit their request to me by email as far in advance as possible.

If you have questions or concerns about your request, you may contact EOCR at UVAEOCR@virginia.edu or (434) 924-3200 or visit their Religious Accommodations webpage for additional information. Please note that receiving an accommodation does not relieve you of your responsibility to complete any coursework you miss as a result of the accommodation.

Reporting an Incident
Just Report It is the University's online system for reporting:

- Sexual and Gender-Based Harassment and Violence
- Discrimination, Harassment, and Retaliation
- Hazing
- Clery Act Compliance (by CSAs)
- Interference with Speech Rights
- Youth Protection
- Preventing & Addressing Threats or Acts of Violence

You may access Confidential Resources if you wish to discuss a concern or incident without reporting to the University. Information you share with Confidential Resources will not be disclosed to University officials or any other person except in extremely limited circumstances. For more information on reporting an incident, visit this website.

Student Mental Health and Wellbeing

The University of Virginia is committed to advancing the mental health and wellbeing of its students, while acknowledging that a variety of issues directly impact your academic performance. Residential MSDS students may access the School of Data Science Embedded Psychotherapist Beth Holt Wright, LCSW by:

- Scheduling online through the Healthy Hoos Portal or
- Emailing mdj7wf@virginia.edu or
- Calling CAPS at 434-243-5150. Notify the receptionist that you are enrolled in the School of Data Science. If you need expedited / urgent support, consider a walk-in appointment at the main CAPS location on the 4th floor of the Student Health and Wellness Center.

If you or someone you know is feeling overwhelmed, depressed, and/or in need of support, contact the CAPS Care Managers at CAPSCareMgrs@virginia.edu.

For help finding a community therapist, visit the Community Referrals page through CAPS.

Additional Resources

The School of Data Science Office of Student Support can help find resources for students experiencing Emergency Needs. Please contact them at SDS_OSS@virginia.edu.